

Candidate Number _____

Fraction _____

2019 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions**First Trial Session A**

I. Restriction enzyme cleavage site detection of plasmid composition

II. Gene function estimation

All the equipment and reagents needed for the experiment are on the table. Please check them according to the list below.

If anything is missing, please raise your hand and let the judges know. After the experiment, please put away all used equipment neatly.
together.

Experimental equipment:

Equipment	Quantity	Pharmaceuticals and materials	Quantity (μl)
1 p20 microdispenser,			
1 sterile water marker, 1 p20 microdispenser pipette tip timer, 1			1 tube (100)
microcentrifuge tube, 1 pair of scissors, 1 microcentrifuge rack, 1 can of glue, 1 bottle			30 sticks
of the following medications placed on ice, 1 5X (multiplied) restriction			6
enzyme reaction buffer, 1 tube (18).			1
Mini centrifuge,			
numbered centrifuge tubes, 1 water bath float, 1 tube of EcoRI restriction enzyme (10).			
Numbered petri dishes (for holding film), 1 plastic A, 1 tube (4)			
Electrophoresis tank (including electrophoresis film), 1 plasmid B, 1 tube (4)			
Water bath (37°C) Common mass C1 tube (4)			
DNA film imaging system common DNA injection dye (labeled: Dye) 1 tube (16)			
		Ruler-labeled DNA (labeled: M) 1 tube (24)	

Please note:

1. Please confirm that your candidate number is correct; if it is incorrect, please raise your hand and ask the teaching assistant for assistance.
2. Once the medicines and equipment on the table are used up, they will not be replenished.
3. Please use public instruments according to the instructions.
4. This exam paper (including the cover and the question paper) consists of 6 pages and must be returned in its entirety when submitting the exam.
5. This exam includes Practice A and Practice B, with a total time limit of 90 minutes.
6. Please answer the questions on this paper. You may write your answers on the back of the questions, but please indicate the question number.

Practical Exercise A (50 points)

Background information:

To determine the function of a specific gene, one common method is to knock out the gene and then examine its structure.

By observing the defects in the knockout mutant, we can infer the possible function of the gene. However, gene knockout sometimes...

It may not be feasible, for example: it may not be easy to implement in some organisms, or the gene that you want to remove may be an essential gene.

If an essential gene is removed, the organism will die. Without obtaining a mutant with the gene removed, it is impossible to perform functional analysis on it.

The workaround is to construct conditional mutations, which prevent the gene from being expressed under specific conditions or at specific times.

Alternatively, the protein being expressed can be disabled, and its phenotype can then be examined. In this way, a mutant can be obtained, and the shutdown can be observed.

This gene deletion can cause problems. For example, the *TEM1* gene in budding yeast is an essential gene that encodes a protein.

Its function is to ensure that during mitosis, the spindle fibers are aligned correctly so that the budding daughter cell receives a nucleus.

Only then will cytoplasmic division occur, separating the cell from the mother cell. Scientists want to investigate whether *TEM1* is also necessary for meiosis.

A conditional mutation of *TEM1* was then constructed, and the promoter of the *CLB2* gene was inserted before the coding region of *TEM1*.

(Promoter, denoted as "*pCLB2*"). This promoter regulates the transcription of the *CLB2* gene, ensuring that *CLB2* is only transcribed in mitotic cells.

It exhibits its function during mitosis but not during meiosis. Therefore, the constructed *pCLB2-TEM1* will only be observed during mitosis.

This phenotype does not occur during meiosis. The two normal *TEM1* alleles originally present in yeast cells are replaced with...

The *pCLB2-TEM1* allele can maintain the normal budding reproduction of this conditional mutant, while meiosis is induced.

At the same time, it is possible to examine whether the effect of shutting down *TEM1* on meiosis and spore formation is any.

To construct the *pCLB2-TEM1* allele, a 2465 bp gene containing the *TEM1* gene was first constructed using PCR and restriction enzymes.

The BamHI-PvuII fragment was selected and colonized on plasmid R904 (Figure 1) to obtain plasmid T770 (Figure 1). Next, PCR and restriction were performed...

The enzyme inserts a 1006 bp SalI-SalI fragment containing *pCLB2* before the *TEM1* gene coding region on plasmid T770, resulting in...

The *pCLB2-TEM1* on plasmid T771 was then transferred to other plasmids, and yeast culture was carried out.

In the cell transformation experiment, the *TEM1* allele on the chromosome was replaced with the *pCLB2-TEM1* allele. This ultimately induced...

Meiosis was performed on the *pCLB2-TEM1* homozygous mutant, and the spore production was examined.

Question 1: Please use restriction enzyme site detection to determine which of the three plasmids (A, B, and C) in the material is plasmid R904.

What is plasmid T770? What is plasmid T771?

Question II: Please analyze and infer the function of the *TEM1* gene in meiosis based on the data provided in the question.

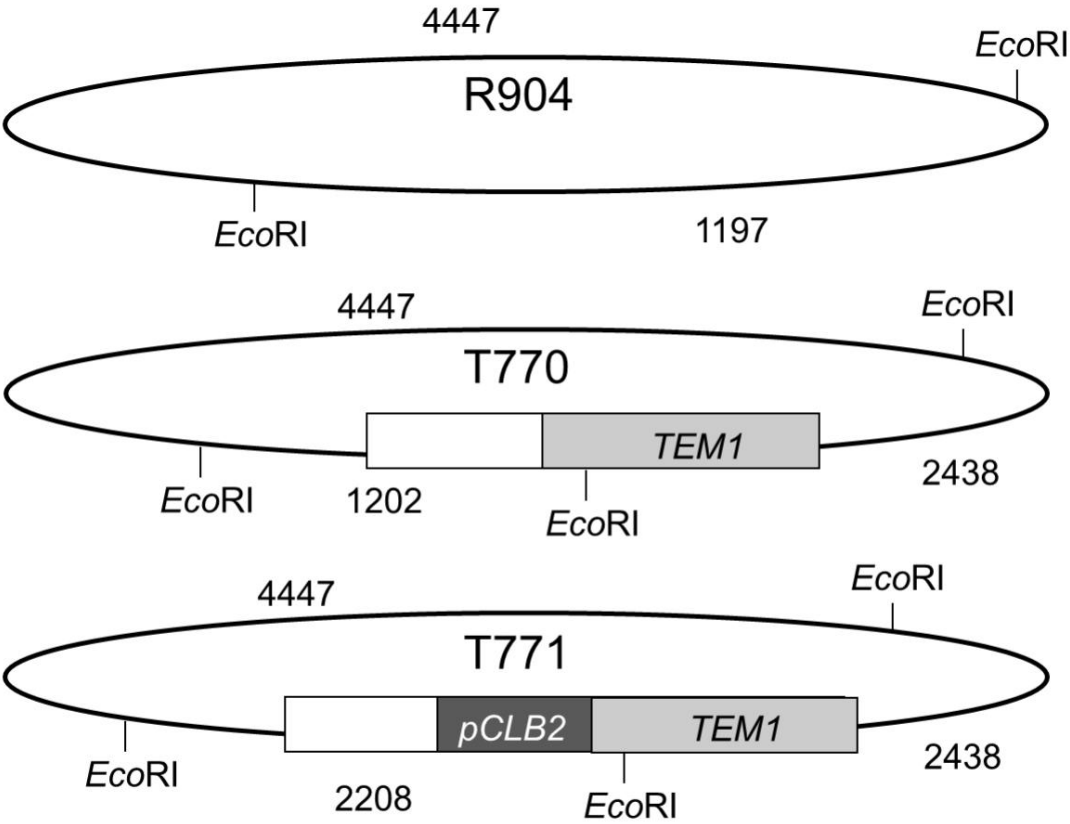


圖1. 質體 R904、T770、T771上 *EcoRI* 限制酶切位及各切位間距離(單位：bp)

Question 1:

Experimental procedures and results: (Procedures worth 5 points)

Q1 cleaves plasmids A, B, and C with the restriction enzyme *EcoRI*, respectively. The reaction mixture should contain 1X [a specific enzyme name - likely a specific enzyme name] during the restriction enzyme reaction.

(Multiple) concentration of restriction enzyme reaction buffer; in single plasmid cleavage reactions, 8 units of restriction enzyme are used, while the experimental materials contain...

The concentration of *EcoRI* is 4 units/ μ l. Based on these conditions, fill in the restriction enzyme cleavage reaction design (for single plasmids) in the table below.

(Preparation quantity for cutting). (6 points)

Formulated into fractional DNA		5X (times) concentration of limiting enzyme reaction buffer	<i>EcoRI</i> sterile water	total volume	
Usage (μ l)	2				20

1. Prepare a mixture sufficient for restriction enzyme reactions of three plasmids according to the design in Q1: Take an empty microcentrifuge tube and prepare the mixture aseptically.

Add water, 5X (fold) concentration reaction buffer, and restriction enzyme *EcoRI* in that order, using a microdispenser, to 3.5g of each from the Q1 table.

When using this microcentrifuge tube to aspirate different solutions, replace the pipette tip and use a microdispenser to aspirate and mix them.

2. Take three more empty microcentrifuge tubes and label them A, B, and C with a marker. Use a microdispenser to dispense the contents of the tubes from step 1.

Take 18 μl of the restriction enzyme reaction mixture and add it to tubes A, B, and C respectively. No pipette tip needs to be changed for this dispensing step.

3. Using a microdispensing device, aspirate 2 μl of DNA from plasmid A, plasmid B, and plasmid C respectively, and add the corresponding restriction enzyme reaction mixture.

The mixture should be aspirated and mixed using a microdispenser in tubes A, B, and C. Each time a different sample is aspirated, a clean dispensing device should be used.

Unused straw tips.

4. Centrifuge the mixture in each tube to the bottom of the tube using a mini centrifuge. When centrifuging, please make sure to place the centrifuge tubes evenly in the centrifuge.

5. Place the centrifuge tubes containing the reaction mixture on a numbered water bath float and in a 37°C water bath for 35 minutes (using a timer).

(Timed), use the waiting time to perform either Practice B or Practice A in Question II.

6. When the 35-minute reaction time is over, retrieve your sample from the water bath.

7. Using a microdispenser, add 4 μl of DNA and dye to each reaction sample, mixing with the dispenser. Change to a fresh aspirator each time.

Tube tip.

8. Using a microdispenser, take **20 μl** of the scale-marked DNA and **20 μl** of the reaction sample containing the added dye, respectively, and dispense according to the scale markings.

Add the DNA, tube A, tube B, and tube C in that order, from left to right, into the sample wells of the electrophoresis gel. Add the DNA gently.

Do not allow the sample to overflow from the trough. Replace the pipette tip each time.

9. Raise your hand to signal the teaching assistant, and follow the teaching assistant's instructions to set up and connect the power supply for electrophoresis for 20 minutes.

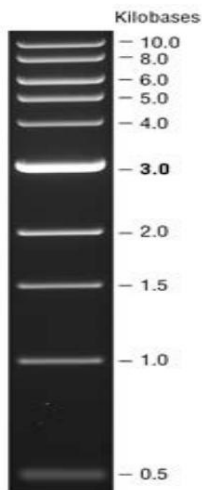
10. After electrophoresis is complete, turn off the power, carefully remove the film, place it on a numbered culture dish, and take it to the film imaging area.

Give the lab assistant to take a photo and print it out. After obtaining the result image, paste the image in the blank space of Q2 and answer the questions based on the image.

Q2 Restriction enzyme reaction and electrophoresis results (12 points)

(base pairs)

Please cut out the electrophoresis result image and glue it here .



Q3. Based on the electrophoresis results and the standard diagram of DNA separation patterns (top left, unit: kb), determine plasmids A, B, and C.

What is R904? What is T770? What is T771? **(9 points)**

Plasmid A:

plasmid B:

plasmid C:

Question 2:

Q4 In diploid yeast, the presence of just one normal *TEM1* allele is sufficient for the cell to undergo budding and meiosis.

The cell divides and produces spores. A heterozygote with the genotype *TEM1/tem1* is currently present. *TEM1* represents a functional normal

Alleles, while *tem1* represents the nonfunctional knockout allele. If this heterozygote is induced to undergo meiosis and produce 4

Given four haploid spores, considering only the *TEM1* genotype, what should be the survival rate of these four spores? Why?

Huh? **(6 points)**

Q5 If *pCLB2-TEM1* can indeed function normally during mitosis and budding in yeast, then the genotype will be...

After the heterozygote of *TEM1/ pCLB2-TEM1* induced meiosis and produced 4 haploid spores, considering only *TEM1*...

Given the genotype, what would be the survival rate of these four spores? Why? **(6 points)**

Q6 To investigate whether the *TEM1* gene (expressing the Tem1 protein) is involved in meiosis and sporulation, a gene structure was first constructed.

Homozygous mutants of type pCLB2-TEM1/pCLB2-TEM1 were induced to undergo meiosis, and then treated with a drug.

benomyl interferes with spindle fiber formation, causing some cells to produce shorter or abnormally oriented spindles. Finally, the production process is examined.

The conidial process. Results are as follows:

Genotype (treatment conditions)	Spore production efficiency (%)	Spore survival rate (%)
TEM1/TEM1 (DMSO)	50.8	75.6
TEM1/TEM1 (benomyl)	39.1	74.3
pCLB2-TEM1/pCLB2-TEM1 (DMSO)	59.2	72.9
pCLB2-TEM1/pCLB2-TEM1 (benomyl)	59.2	59.7

* Benomyl must be dissolved in DMSO solvent before being applied to cells.

Based on these experimental results, discuss the possible functions of the yeast Tem1 protein in meiosis and sporulation. **(6 points)**

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2019 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions

First Trial Session B

Biochemistry Experiment

Note: Operation time is limited, please make good use of the time waiting for the experiment to take effect.

• All necessary equipment and reagents for the experiment are on the table. Please check them against the list below. If anything is missing...

Please raise your hand to inform the examiners. After the experiment, please clean the used equipment and put it away neatly.

Experimental equipment and reagents:

equipment	Quantity of	experiments and experimental materials	
Micro-dispenser P20	1 reaction	buffer	2 tubes
Micro-dispenser P200	1 piece	(0.1M Tris-HCl, 0.5 M	
Micro-dispenser P1000	1 piece	MgCl ₂ , 0.01 M CaCl ₂ , pH 8)	
Micropipette tip 20~200 µL	1 box of	deionized water	1 tube
Micropipette tip 200~1000 µL	1 box of	diatom sample A	1 tube
microcentrifuge tubes	10.	Diatomaceous earth sample B	1 tube
centrifuge	1 unit of	reaction termination solution	1 tube
Micro colorimeter	1 plate	(0.25 M EDTA, 0.1 M Tris-HCl, pH 8)	
Micro colorimeter	Shared	Placed on ice	
Oil-based brush	1	Alkaline phosphorylase substrate (13.5 mM p-nitro-phenylphosphate)	1 tube
Waste liquid cup	1 control	group sample A	1 tube
Vibrating Mixer	1 control	group sample B	1 tube
timer	1 unit		
stapler	1 unit		
Centrifuge tube rack	1		

Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. This exam paper (including the cover and the question paper) has a total of 4 pages and must be returned in its entirety when submitting the exam.
3. This exam includes Practice A and Practice B, with a total time limit of 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back of the paper, but please indicate the question number.

I. Experiment Theme: Using biochemical experiments to detect changes in the marine ecological environment

Algae are important primary producers in the marine environment, and diatoms are among the most important phytoplankton, providing over 40% of marine primary productivity and sequestering over 20% of global carbon dioxide. Therefore, in the field of marine environmental ecology research, the abundance and growth status of diatoms in a given area is a crucial ecological observation project, and economically, it can also be used to predict the future of related marine areas.

Fishery resources, etc.

Algal growth is greatly influenced by changes in environmental nutrients, and the distribution of nutrients in the marine environment is extremely uneven. Sometimes, a region may suddenly become rich in conditions that promote algal growth, resulting in an algal bloom, a phenomenon characterized by massive algal proliferation. If the algae forming the bloom are toxin-producing, it can potentially cause mass mortality of marine life and even food safety issues. Conversely, the ocean may also experience nutrient deficiency, leading to a decline in algal populations in that area. The distribution of nutrients in the ocean is constantly affected by ocean currents, climate, the rise and fall of species within the ecosystem, and human activities. Therefore, regular observation of marine ecosystems can help us understand patterns of environmental change, especially the impact of human activities on the ocean.

What kind of interference did it cause?

Phosphate is an essential nutrient for algal growth and can be used to produce phospholipids and nucleic acids.

Phosphate is a major component of algal cells. When there is insufficient phosphate in the environment, algal growth is inhibited. At this time, most algae will begin to produce "alkaline phosphatase," using this enzyme to break down organic phosphorus in the environment (derived from components released by the death of other cells), and then reuse the resulting phosphate. In this experiment, students will act as marine ecologists, using the characteristic of alkaline phosphatase in algae when they are deficient in phosphorus, to analyze two tubes of marine diatom samples using biochemical methods, detecting the alkaline phosphatase activity in the two tubes, and then judging whether the algae in the samples are deficient in phosphorus based on the experimental results.

The growth environment of acid salts.

II. Experimental Procedure: (10 points)

2-1 Algal Sample Processing

ŸNote : The experimental procedure is a double-repetition operation.

1. Carefully observe the two algae samples (tubes A and B) to see if there are any differences in appearance. We need one of them.

For subsequent experiments, please consider how to take the sample. _____

2. Take 1 mL from each of the two algal samples and add them twice to four different microcentrifuge tubes, following the instructions...

The groups in the table are labeled as 1-1 to 2-2.

3. Place the microcentrifuge tubes containing the sample symmetrically into a benchtop centrifuge and centrifuge for 2 minutes at room temperature.

(Please raise your hand before centrifuging so the lab assistant can check.)

4. After centrifugation, use a micro-dispenser to aspirate the supernatant into a waste cup, leaving only the precipitate (diatoms).

(Regarding the cellular portion), please consider how to remove the supernatant without aspirating the cells. _____

2-2 Prepare each group of reactions

1. Using a micro-dispenser, take 290 µL of the "reaction buffer" and add it to step 4 of section 2-1 above.

In a microcentrifuge tube, the diatom cell precipitate was shaken to mix and then allowed to stand.

2. Control Group Preparation: To ensure that the reagents used in the experiment are safe, a control group needs to be designed.

In this experiment, purchased "shrimp alkaline phosphate solubilizer" (SAP) will be used as the positive control group.

Then, the "reaction buffer" was used as the negative control group (blank group). However, due to the preparation process in the laboratory...

An error occurred: the sample labels for the positive and negative control groups were lost. However, we have determined that there is one positive and one negative control tube in each of the two tubes.

Therefore, the two tubes had to be randomly labeled as control group A and control group B. Please refer to Table 1 for further details.

Take four new microcentrifuge tubes, prepare the standard and control group samples, mix well and let stand.

Table 1. Experimental Sample Configuration

reaction tube Added ingredients	1-1	1-2	2-1	2-2	3-1	3-2	4-1	4-2
algal samples	fine sediment Cell A	fine sediment Cell A	fine sediment Cell B	fine sediment Cell B				
control group sample product	0	0	0	0	control group A 10	control group A 10	control group B 10	control group B 10
Reaction buffer liquid	290	290	290	290	280	280	280	280
total (µL)	290	290	290	290	290	290	290	290

2-3 Perform phospholytic enzyme activity test

1. Using a microdispenser, take 10 μL of alkaline phosphorylase matrix and add it to each of the prepared reaction mixtures.

In the sample, mix well and allow to stand at room temperature for 20 minutes to react. Please consider how to mix the ingredients more effectively.

good.

2. After the reaction is complete, add 30 μL of reaction terminator to each tube.
3. Centrifuge for 2 minutes (please raise your hand for the lab assistant to check before centrifuging).
4. After retrieving the sample, take out the micro-volume colorimetric disk and add 200 μL of deionized water to the first compartment as a blank.

Group the samples, then add reaction samples 1-1 to 4-2 from step 2-2 in sequence. Write your personal information on the edge of the colorimetric disc.

After the exam number is issued, the results will be handed over to the experimental teaching assistant to measure the absorbance at 410 nm and recorded in the experimental results.

III. Experimental Results (20 points)

1. Observe algae samples A and B. What are the differences between the two samples in terms of appearance?

2. Record the 410 nm absorbance values in the table below.

reaction tube	Deionized water	1-1	1-2	2-1	2-2	3-1	3-2	4-1	4-2
OD410									

Please staple the 410 nm absorbance value printed by the teaching assistant to the end of the title.

3. Based on the experimental results, which group of algae grows in a phosphorus-deficient environment?

4. Based on the experimental results, which group is the positive control group?

IV. Questions and Answers **(20 points)** 1. If

this experiment is conducted by a team of oceanographic research vessels collecting seawater from a designated location, after three weeks...

After the voyage, the enzymes were brought back to a land-based laboratory for the aforementioned biochemical experiments to analyze their activity. Would you suggest...

Which of the following methods should be used to preserve and bring back the sample after collection? (A) Add formalin and store at room temperature. (B) Sterilize at

high temperature and then freeze. (C) Place directly in a 4-degree refrigerator. (4 points) What are your reasons for making this suggestion? (4 points)

2. If you want to determine how many diatom cells are in each of the collected algal samples A and B, what would you do?

What measurement method will be used, and what kind of equipment will be needed? (4 points)

3. If the above experimental steps are followed, the measured OD410 values for the two samples are: Group A: 0.4

The value is 0.5 compared to Group B. When the values are very close, it can be concluded that the algae in Group B live in a more suitable environment.

Is the car deficient in phosphorus? Please explain why. (4 points)

4. Which of the following locations is most likely to yield algae samples with high alkaline phosphatase activity? (A) Freshwater estuary (B) The sea area where "Mazulan

tears" appear (C) Open ocean waters without ocean currents (4 points)